Hunor Csala

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Education

University of Utah **2022-2025**
PhD in Mechanical Engineering *Salt Lake City, UT, USA*
Advisor: Dr. Amir Arzani
Thesis: Exploiting low-dimensionality for data-driven cardiovascular flow modeling

Budapest University of Technology and Economics **2019-2021**
MS in Mechanical Engineering *Budapest, Hungary*
Advisor: Dr. Dániel Gyürki
Thesis: Emerging fractal patterns in blood flood inside intracranial aneurysms using the Lattice Boltzmann method (LBM)

Budapest University of Technology and Economics **2015-2019**
BS in Mechanical Engineering *Budapest, Hungary*
Advisor: Dr. László Bencsik
Thesis: Model predictive control (MPC) of human balancing on a balance board

Research Interests

Core Areas: Scientific Machine Learning (SciML), Data-Driven Modeling of Physical Systems, Computational Fluid Dynamics (CFD), Theoretical & Applied Fluid Mechanics, Cardiovascular Flow Modeling and Biomechanics, Control & Optimization

Experience

Oak Ridge National Laboratory **June 2025 –**
Postdoctoral Research Associate - Computational Sciences and Engineering Division *Oak Ridge, TN, USA*

- Worked on vision transformer–based foundation models (200M+ parameters) for turbulence and fluid mechanics, trained on 20+ datasets with diverse physics. Trained and fine-tuned on 8 multi-node allocations (64 GPUs total) using PyTorch Distributed Data Parallel and sequence parallelism to handle large 3D spatiotemporal inputs.
- Designed transformer-based reduced-order models for autoregressive prediction of plasma edge dynamics in nuclear fusion reactors; achieved ¡6% error over 1000-step rollouts using unseen input control actuation signals. Performed multi-GPU distributed training on the PSC Bridges-2 system (up to 8 GPUs)
- Conducted large-scale distributed training on multiple GPUs using the OLCF Frontier supercomputer

Scientific Computing and Imaging Institute, University of Utah **Aug 2022 – May 2025**
Graduate Research Assistant *Salt Lake City, UT, USA*

- Performed large-scale CFD simulations of patient-specific cardiovascular flows using the finite element method in SimVascular; ran MPI-parallel jobs on up to 128 CPU cores with meshes up to 15 million elements on the University of Utah Center for High Performance Computing (CHPC), managing domain decomposition, job submission, and scaling
- Compared multiple nonlinear dimensionality reduction techniques for unsteady fluid flow modeling
- Developed supervised and unsupervised autoencoder-based machine learning frameworks for denoising and imputing 4D-Flow MRI cardiovascular flow data, improving the reconstruction accuracy by 25% compared to traditional methods
- Worked on a 1D blood flow surrogate model using physics-constrained neural ODEs

Los Alamos National Laboratory **Summer 2023 & 2024**
Graduate Research Assistant - Computing and Artificial Intelligence Division *Los Alamos, NM, USA*

- Developed a new physics-constrained data-driven method using neural partial differential equations
- Implemented neural PDE in Julia language, using differential programming with Zygote and DiffEqFlux
- Neural PDE approach has 3-5 times smaller error for flow rate, area and pressure than state-of-the-art 1D FEM for blood flow in stenosed arteries using elastic walls at similar computational cost
- Automated meshing, FSI simulation setup, and batch execution for up to 100 parameterized geometries using Bash and Python scripting, enabling reproducible high-fidelity blood flow data generation

Northern Arizona University **Aug 2021 – Aug 2022**
Graduate Research and Teaching Assistant *Flagstaff, AZ, USA*

- Used autoencoders and manifold learning for dimensionality reduction of CFD data, trained on NAU's Monsoon cluster

Furukawa Electric Institute of Technology **Jun 2018 – Aug 2021**
Simulation Engineer Intern *Budapest, Hungary*

- Worked on CFD simulation of keyhole mode laser welding using OpenFOAM
- Developed a corrected ray tracing model for radiation modeling in OpenFOAM using a custom radiation module
- Worked on CFD simulation of an MOCVD reactor in OpenFOAM

Publications

***H.Csala**, S. De Pascuale, P. Laiu, J. Lore, J. S. Park, P. Zhang: Autoregressive long-horizon prediction of plasma edge dynamics, Nuclear Fusion 66 (6), 066013* *2026*

***H.Csala**, A. Arzani: Decomposed Sparse Modal Optimization (DESMO): Application to unsteady flows, International Journal of Heat and Fluid Flow 117, 110124* *2026*

***H. Csala**, A. Mohan, D. Livescu, A. Arzani: Physics-constrained coupled neural differential equations for one dimensional blood flow modeling, Computers in Biology and Medicine 186, 109644* *2025*

*A. Pashaei Kalajahi, **H. Csala**, Z. Mamun, S. Yadav, O. Amili, A. Arzani, R. D'Souza: Input Parameterized Physics Informed Neural Networks for Denoising, Super-resolution, and Imaging Artifact Mitigation in Time Resolved Three Dimensional Phase-Contrast MRI, Engineering Applications of Artificial Intelligence 150, 110600.* *2025*

***H. Csala**, O.Amili, R. D'Souza, A. Arzani: A comparison of machine learning methods for recovering noisy and missing 4D flow MRI data, International Journal for Numerical Methods in Biomedical Engineering 40 (11)* *2024*

***H. Csala**, A. Mohan, D. Livescu, A. Arzani: Modeling Coupled 1D PDEs of Cardiovascular Flow with Spatial Neural ODEs, NeurIPS Machine Learning for the Physical Sciences Workshop* *2023*

***H. Csala**, S. Dawson, A. Arzani: Comparing different nonlinear dimensionality reduction techniques for data-driven unsteady fluid flow modeling, Physics of Fluids 34 (11)* *2022*

Presentations

***H. Csala**, P. Zhang: Finetuning MATEY foundation model for diverse fluid systems and deep learning tasks*
Artificial Intelligence for Robust Engineering & Science (AIRES) 7 - Oak Ridge, TN, USA Aug 2026

*S. De Pascuale, A. Diaw, **H. Csala**, P. Zhang, P. Laiu, J. Lore, J.S. Park: Using Transformational Machine Learning Techniques to Enable Controlled Predictions of Tokamak Plasma Exhaust*
International Conference on Data-Driven Plasma Science - Kiel, Germany Aug 2026

***H. Csala**, P.Zhang: MATEY - Multiscale Adaptive Foundation Model for Spatiotemporal Systems*
Oak Ridge Postdoctoral Association 14th Annual Research Symposium - Oak Ridge, TN May 2026

*P. Zhang, **H. Csala**, S. De Pascuale, P. Laiu, J. Lore, J.S. Park: Autoregressive long-horizon prediction of plasma edge dynamics*
Invited Seminar - Lawrence Livermore National Laboratory (LLNL) - Virtual April 2026

***H. Csala**, S. De Pascuale, P. Laiu, J. Lore, J.S. Park, P. Zhang: MATEY for long-horizon prediction of plasma edge dynamics*
National Artificial Intelligence Research Resource (NAIRR) Annual Meeting - Arlington, VA, USA Mar 2026

*S. De Pascuale, P. Sar, J. Restrepo, G. Staebler, J.M. Park, A. Diaw, M. Cianciosa, **H. Csala**, P. Zhang: Parameter Estimation of Machine Learning Models for Tokamak Pulse Simulation using Adaptive Methods*
APS Division of Plasma Physics - DPP, Long Beach, CA, USA Nov 2025

***H. Csala**, S. De Pascuale, P. Laiu, J. Lore, P. Zhang: MATEY for Fusion: Fast Surrogates for Plasma Edge Dynamics with Vision Transformers*
Artificial Intelligence for Robust Engineering & Science (AIRES) 6 - Oak Ridge, TN, USA Sep 2025

***H. Csala**, S. De Pascuale, P. Laiu, J. Lore, P. Zhang: MATEY for autoregressive prediction of plasma edge dynamics (poster)*
ORNL Software and Data Expo (OSDX) 2025 - Oak Ridge, TN, USA Sep 2025

*A. Arzani, **H. Csala**, S. Viknesh: Interpretable reduced-order modeling of fluid flow and dynamical systems with ADAM optimization*
US National Congress on Computational Mechanics - USNCCM, Chicago, IL, USA July 2025

***H. Csala**, A. Mohan, D. Livescu, A. Arzani: Physics-Constrained Coupled Neural Differential Equations: Application to 1D Blood Flow Modeling*
APS Division of Fluid Dynamics - DFD, Salt Lake City, UT, USA Nov 2024

***H. Csala**, A. Mohan, D. Livescu, A. Arzani: Spatial Neural ODEs for Modeling Blood Flow in Stenosed Arteries with Deformable Walls*
World Congress on Computational Mechanics - WCCM/PANACM, Vancouver, BC, Canada Jul 2024

***H. Csala**, A. Mohan, D. Livescu, A. Arzani: Modeling Coupled 1D PDEs of Cardiovascular Flow with Spatial Neural ODEs (poster)*
NeurIPS, Machine Learning for the Physical Sciences, New Orleans, LA, USA Dec 2023

***H. Csala**, O. Amili, R. D'Souza, A. Arzani: Machine Learning-Based Denoising and Imputation of Unsteady Cardiovascular Flow Data*
APS Division of Fluid Dynamics - DFD, Washington DC, USA Nov 2023

***H. Csala**, O. Amili, R. D'Souza, A. Arzani: Enhancing Corrupt Cardiovascular Flow Data with Machine Learning*

Summer Biomechanics, Bioengineering, & Biotransport Conference - SB3C, Vail, CO, USA — Jun 2023

***H. Csala**, S. Dawson, A. Arzani: Manifold learning and deep autoencoders for nonlinear embedding of unsteady fluid flows*
APS Division of Fluid Dynamics - DFD, Indianapolis, IN, USA — Nov 2022

***H. Csala**, S. Dawson, A. Arzani: Comparing Different Nonlinear Dimensionality Reduction Techniques for Data-Driven Unsteady Fluid Flow Modeling (poster)*
Summer School on Reduced Order Methods in Computational Fluid Dynamics, SISSA, Trieste, Italy — Jul 2022

***H. Csala**, S. Dawson, A. Arzani: Comparing Different Nonlinear Dimensionality Reduction Techniques for Data-Driven Unsteady Fluid Flow Modeling*
US National Congress on Theoretical and Applied Mechanics - USNCTAM, Austin, TX, USA — Jun 2022

Teaching Experience

University of Utah — **Salt Lake City, UT, USA**
Graduate Teaching Assistant — *2023–2024*

- Delivered guest lectures for undergraduate *Fluid Mechanics* and graduate-level *Scientific Machine Learning*.
- Facilitated graduate journal clubs focused on dissecting peer-reviewed literature and computational methodologies.
- Served as Teaching Assistant and Grader for core courses: *Fluid Mechanics* and *Numerical Methods for Engineering Systems*.

Northern Arizona University — **Flagstaff, AZ, USA**
Recitation Instructor — *2021*

- Primary instructor for two undergraduate *Thermodynamics* recitation sections totaling 60 students.
- Designed and led weekly interactive blackboard problem-solving sessions, integrating digital quizzes and visual flow animations to reinforce core concepts.

Budapest University of Technology and Economics — **Budapest, Hungary**
Undergraduate Teaching & Laboratory Assistant — *2017–2018*

- Guided freshman engineering students through experimental fluid dynamics and thermal system labs (*Introduction to Mechanical Engineering*).
- Led hands-on data analysis and hypothesis testing sessions in Excel for *Statistics for Mechanical Engineers*.

Manuscripts Under Preparation

Connections between classical numerical methods for differential equations and modern deep learning - A review — *2026*

Towards Full-Field Hemodynamic Reconstruction from One-Point Phase Contrast MRI via Input-Parameterized Physics-Informed Neural Networks — *2026*

IGNOS: An Intelligent Generative Neural foundation mOdel for fusion energy Sciences — *2026*

Software

*Zhang, Pei, **Csala, Hunor**, Prokopenko, Andrey, Yin, Junqi, Gounley, John, Laiu, Ming Tse P., Meena, Muralikrishnan Gopalakrishnan, Palash, Mijanur, & Sankaran, Ramanan. (2026, January 26). MATEY. [Computer software]. https://github.com/ORNL/MATEY. https://doi.org/10.11578/dc.20260126.3.* — *2026*

Honors and Awards

Finalist for Your Science in a Nutshell Competition - Oak Ridge, TN — ORNL

ORPA 14th Research Symposium - Best Presentation - AI/ML Session I - Oak Ridge, TN — ORNL

Researcher of the Year 2025 - TFES, Department of Mechanical Engineering - Salt Lake City, UT — University of Utah

Graduate Student Travel Assistance 2024 - Vancouver, Canada — University of Utah

WCCM 2024 Travel Award - Vancouver, Canada — USACM

Travel Funding from Associated Students of the University of Utah 2023 - Vail, CO — University of Utah

Presidential Fellowship - Northern Arizona University 2021-2022: $9,000 award - Flagstaff, AZ — NAU

College of Education - Graduate Research Student Travel Award 2022 - Austin, TX — NAU

Campus Mundi Scholarship 2020: Erasmus Study Abroad Program - Naples, Italy — Tempus Public Foundation

Pro Progressio Foundation FETI Research Scholarship 2018-2021 - Budapest Hungary — Pro Progressio Foundation

ELTE Márton Áron Special College - Member 2018-2021 - Szombathely, Hungary ELTE

3rd Prize at International Hungarian Mathematics Competition 2014 - Miercurea Ciuc, Romania NMMV

Service

Journal Referee :

- Physics of Fluids
- Physical Review E
- Physical Review Materials
- Physical Review Research
- International Journal for Numerical Methods in Biomedical Engineering
- Physical Review Letters
- NeurIPS ML4PS Workshop (2023-2025)
- NeurIPS AI4Science Workshop (2025)
- ICML AI4Science Workshop (2026)

Leadership: Oak Ridge Postdoctoral Executive Committee 2026 - Social and Well-being Co-Chair

Member: American Physical Society (APS), National Postdoctoral Association (NPA)

Volunteer: APS DFD 2024, ORPA Research Symposium 2025, 2026, Presidential AI Challenge Judge 2026

Organizer: SCI X 2025

Skills

CFD	OpenFOAM, SimVascular, Palabos, Fluent	**HPC**	Linux, slurm
FEM	FEniCS, ANSYS Mechanical	**ML**	PyTorch, scikit-learn, zygote, pandas, scipy
Languages	Python, Matlab, Julia, C++, Wolfram	**Other**	Latex, git, ParaView, vtk, SolidWorks

References

Amirhossein Arzani, Associate Professor, University of Utah: amir.arzani@sci.utah.edu PhD Advisor

Arvind Mohan, Staff Scientist, Los Alamos National Laboratory: arvindm@lanl.gov Internship Mentor

Pei Zhang, Staff Scientist, Oak Ridge National Laboratory: zhangp1@ornl.gov Postdoc Advisor

Daniel Livescu, Staff Scientist, Los Alamos National Laboratory: livescu@lanl.gov Internship Mentor